
Wiki3.ai

Search for a World of Local Agents

A distributed, local-first knowledge system where AI can process, organize, cite, and reason over the world's information without one platform owning the corpus, index, or queries.

Thesis: local AI has arrived; local-first discovery has not.



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The Local-First AI Moment

- Capable models now run on laptops, workstations, and private servers.
- Local embedding, reranking, OCR, extraction, and structured-information pipelines are practical.
- Personal agents can work over private files, code, notes, repositories, and organizational data.
- Users increasingly want privacy, offline capability, predictable cost, and ownership of their AI context.

But an agent is only as capable as the information it can discover.

The Missing Layer: Search

Today, an agent needing information outside its local corpus typically calls a cloud search product:

- Exa
- Tavily
- DuckDuckGo
- Google/Bing-derived services
- Proprietary “web research” APIs

That is convenient, but it recreates central dependence at the most important boundary:

Who decides what the agent can find, rank, cite, and remember?

Local RAG Is Solved Enough

For a bounded personal or team corpus, the basic local RAG stack is familiar:

INPUT

Local files / Git / notes / PDFs / databases

PREPARE

Extract → chunk → embed → index

RETRIEVE

Hybrid retrieval → rerank → local or remote LLM

OUTPUT

Answer with local citations

The hard problem begins when an agent needs to search beyond the local corpus.

The Open World Is Already Indexed

Open corpora, searchable indexes, and public query services already exist:

- **Common Crawl:** web archives, CDXJ, and URL indexes, with public index data on S3.
- **OpenAlex:** an open scholarly graph with CC0 metadata through bulk download and an API.
- Wikidata and SPARQL endpoints; OpenStreetMap and Overpass services.
- Government DCAT/CKAN catalogs; institutional repositories, libraries, and archives.
- Public OpenSearch, Elasticsearch, and Solr deployments.
- Domain indexes for code, standards, patents, genomics, and research.

The problem is a lack of a common discovery layer.

Sources: [Common Crawl index server](#) · [CDXJ index](#) · [URL index](#) · [OpenAlex](#)

An Open Index of Search

Build a distributed registry of **searchable resources** for AI agents.

Each entry describes a way to discover and retrieve evidence:

```
{
  "name": "Example Scholarly Corpus",
  "topics": ["science", "papers"],
  "license": "CC0",
  "access": ["bulk-download", "REST-search"],
  "endpoint": "https://example.org/api/search",
  "protocol": "OpenAPI",
  "query_modes": ["keyword", "metadata-filter", "semantic"],
  "provenance": "publisher / crawl / mirror",
  "update_cadence": "daily", "local_mirror": true
}
```

Think DNS + package registry + sitemap + DCAT for agent-searchable knowledge.

Agents Need Capabilities, Not URLs

An agent-readable search capability descriptor should answer:

- What does this resource cover?
- Is metadata and/or content open?
- Anonymous, user-key, or local access?
- Which query modes are supported?
- Are stable IDs and source spans available?
- Can it be mirrored or synchronized?
- What are cost, rate, privacy, and license constraints?
- Is it healthy and trustworthy?

The unit of federation is a search capability descriptor.

A Local-First Retrieval Architecture

Open search registry

Signed descriptors, health, provenance



Local agent discovery planner

Chooses sources using policy and capabilities



Local indexes

Files / Git
Notes / databases
Private corpora

Local mirrors

Common Crawl
OpenAlex
Domain datasets

Remote open endpoints

SPARQL / APIs
Search services
Public indexes

Hybrid retrieval + evidence/provenance bundle

Agent answer with citations

Where OpenSearch Fits

Search infrastructure can expose capabilities directly to agents:

- **Portable manifests** for hosted or self-hosted indexes.
- **Federated querying** across local indexes, mirrors, and public endpoints.
- **Hybrid retrieval** across lexical, vector, metadata, temporal, and access-aware queries.
- **Incremental syndication** through feeds, snapshots, change logs, or content-addressed artifacts.
- **Evidence-grade results** with IDs, spans, timestamps, hashes, and provenance.
- **Local replicas** that cache relevant shards without copying the whole world.
- **Inspectable ranking** that explains why an agent received a result.

The opportunity: make search infrastructure discoverable and composable by agents.



The Question for This Community

How do we build and maintain an open index of search datasets, indexes, and endpoints that local-first agents can discover, query, mirror, and cite?

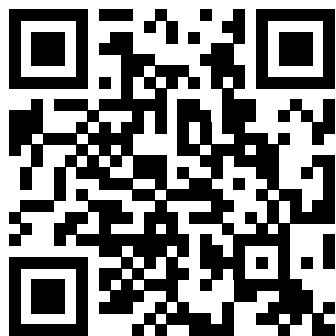
A useful first milestone:

1. A minimal open schema for agent-readable search capabilities.
2. A signed, federated registry with decentralized mirrors.
3. Adapters for OpenSearch, Elasticsearch, Solr, SPARQL, CKAN/DCAT, and Common Crawl-style indexes.
4. A local agent client spanning private data, local mirrors, and public sources.
5. Provenance, licenses, permissions, freshness, and reproducibility as retrieval fields.

If we do not build this layer openly, a few centralized search APIs will mediate the “AI web,” even when models and agents run locally.

Join the Wiki3 conversation

Explore the project and join the Zulip discussion about open search for local agents.



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Wiki3 Zulip invite

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